

Decoding Customer Value: A SQL-Driven Retention Strategy

Executive Summary

This project examined the purchase behaviour of customers of an online direct-to-consumer(D2C) fashion brand in USA, looking to better understand customer value, loyalty, reliance on promotions and long-term retention strategies. Python, SQL and Power BI were used to develop customer-level behavioural metrics and segmentation models for strategic decision making based on data.

The segmentation analysis of the customers showed good separation between value tiers. Overall, VIP customers spent an average of \$84.06 per purchase compared to the average of \$34.50 per purchase of Low Value customers, and they also had a significantly higher purchase frequency (25.42 purchases versus 10.73 purchases) and purchase history (34.61 purchases versus 16.91 purchases). This indicates that customer value is strongly associated with sustained engagement and repeat purchasing behavior rather than single high-value transactions.

Two competing definitions of loyalty were tested. The behavioral loyalty model, which was derived mainly from purchase history and purchase frequency, gave better separation of the customers than the alternative, relationship-based model. When it came to customer value, the loyalty analysis showed that about 84% of VIP customers were in the High Loyalty and Champion segments, indicating a high level of alignment between value and repeated engagement.

Promotional analysis showed low level of customer behavioural differences between the discount users and non-discount users across customer segments. Champion customers using discounts averaged \$59.48 per purchase compared to \$60.50 among non-discount users, while previous purchase behavior remained nearly identical across both groups. The same pattern was noticed in the various product categories and geographical regions. The results indicate that promotions were not very effective at distinguishing customer quality and parts of the current discount policy may be subsidizing customers already having high repeat purchase behaviours.

Using geographic analysis, we were able to identify certain regions (Arizona, Alaska, Pennsylvania, Hawaii, Wyoming, Illinois and Alabama) that were strong in multiple dimensions of customer quality such as customer value, repeat engagement, VIP concentration, organic demand and promotional independence. Product-category analysis revealed that there were relatively few differences between product categories, indicating that customer value is more influenced by behavioral engagement than by just having a category preference alone.

This led to a Selective Promotional Sunset Plan which involves phasing out the blanket discounting of more loyal customers and progressively increasing targeted retention and engagement initiatives for high-potential customer segments.

The overall analysis indicates that long-term customer value is more strongly related to purchase frequency, stronger purchase history and not discount participation. The ideal customer is a High Loyalty or Champion customer who is a VIP.